

Harish Kumar A

Robotics Software Engineer | ROS 2 | Motion Control | Autonomous Systems

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PROFESSIONAL SUMMARY

Robotics Software Engineer with 1 year of industry experience developing, integrating, field-testing, and debugging physical AGV/AMR systems using Python, C++, Linux, and ROS 2. Experience spans autonomous navigation, path planning, closed-loop motion control, perception, sensor integration, and embedded communication, with foundational manipulator work in forward/inverse kinematics, workspace, and singularity analysis.

EXPERIENCE

Software Engineer

Mar. 2026 – Aug. 2026

Lambda Robotics

Kovilpatti, Tamil Nadu

- Developed and field-tested navigation, localization, motion-control, and embedded-communication functions for autonomous AGV/AMR platforms using Python, C++, Linux, and ROS 2.
- Integrated LiDAR, cameras, IMUs, encoders, motor drivers, Raspberry Pi, and ESP32 controllers across physical robots, connecting perception and robot-state inputs with motion-control and diagnostic workflows.
- Built diagnostic telemetry and systematically resolved failures involving serial communication, heading drift, motor direction, sensor accuracy, camera connectivity, networking, and power sequencing during real-robot testing.
- Developed modular PyQt6 HMI components for task assignment, mission monitoring, robot state, and bidirectional Python-JavaScript communication using Qt WebChannel.

Software Engineer Intern

Sep. 2025 – Feb. 2026

Lambda Robotics

Kovilpatti, Tamil Nadu

- Contributed to robotics prototypes across autonomous navigation, computer vision, embedded systems, sensors, motors, electronics integration, and physical robot validation.
- Developed and tested a Wi-Fi-controlled obstacle-avoiding robot and supported early AGV integration and system-level debugging.

PROJECTS

Vision & IMU-Based Autonomous Guided Vehicle | *Python, C++, OpenCV, AprilTags, ESP32*

Mar. 2026 – Aug. 2026

- Built and field-tested a differential-drive AGV combining AprilTag localization, IMU heading feedback, encoder inputs, A* path planning, lateral course correction, and closed-loop turning on a physical robot.
- Implemented differential-drive motion calculations, velocity control, jerk-limited acceleration, predictive braking, and Raspberry Pi-ESP32 communication for motion commands and localization correction.
- Instrumented runtime telemetry for heading, gyro rate, lateral error, wheel velocity, pulse rate, acceleration, and travel distance; achieved stable operation at approximately 0.245 m/s and 40 wheel RPM.

Six-Axis Robotic Arm – Kinematics & Cycloidal Drive Study | *FK/IK, Workspace, FreeCAD*

Feb. 2026

- Contributed to the conceptual development of a six-axis robotic arm for automated dishwashing, studying forward/inverse kinematics, joint coordinate systems, workspace, singularities, joint motion, and critical load points.
- Evaluated compact high-reduction transmission approaches and designed cycloidal gear components for robotic-arm joints, connecting kinematic constraints with workspace, load, and joint-design considerations.

Autonomous Mobile Service Robot | *ROS 2, SLAM, LiDAR, Depth Camera, YOLO*

May 2026

- Integrated LiDAR and depth-camera inputs with ROS 2 SLAM-based mapping and localization for autonomous indoor navigation in a customer-facing service-robot environment.
- Applied YOLO-based person and obstacle detection to support environment-aware speed reduction and stopping behavior around people and surrounding objects.

AGV Vehicle Control Unit & Power Management | *ESP32, Raspberry Pi, GPIO, Linux, systemd*

Mar. 2026 – May 2026

- Designed an ESP32-Raspberry Pi control architecture with relay-based power sequencing, power-hold logic, graceful Linux shutdown, and GPIO handshaking for coordinated robot startup and shutdown.
- Configured systemd services to initialize robot software at boot and coordinate application readiness with the embedded power-control workflow.

TECHNICAL SKILLS

Programming: C++, Python, C, Java

Robotics & Control: ROS 2, Motion Control, A* Path Planning, Autonomous Navigation, Localization, SLAM Integration, Differential-Drive Kinematics, Closed-Loop Turning, Velocity Control

Manipulation Fundamentals: Forward Kinematics, Inverse Kinematics, Joint Coordinate Systems, Workspace Analysis, Singularity Concepts

Perception & Sensors: OpenCV, YOLO, AprilTags, LiDAR, Depth Cameras, IMU, Encoders, Sensor Fusion

Systems & Tools: Linux, Docker, Webots, Git, GitHub, systemd, PyQt6, Raspberry Pi, ESP32, UART, I2C, GPIO, SSH

EDUCATION

Dr. Mahalingam College of Engineering and Technology

Pollachi, Tamil Nadu

B.Tech in Artificial Intelligence and Data Science; CGPA: 8.14/10

2022 – 2026